



Foundations Of Machine Learning

Speech Emotion Recognition

**Deliverable 3: Model Compression Improvements
for Speech Emotion Recognition**

▮ Deliverable 2 vs Deliverable 3 Improvement ▮

RAVDESS DATASET

DELIVERABLE	Before Pruning	After 40% Pruning
Deliverable 2	73.30%	62.20%
Deliverable 3	77.40%	73.60%

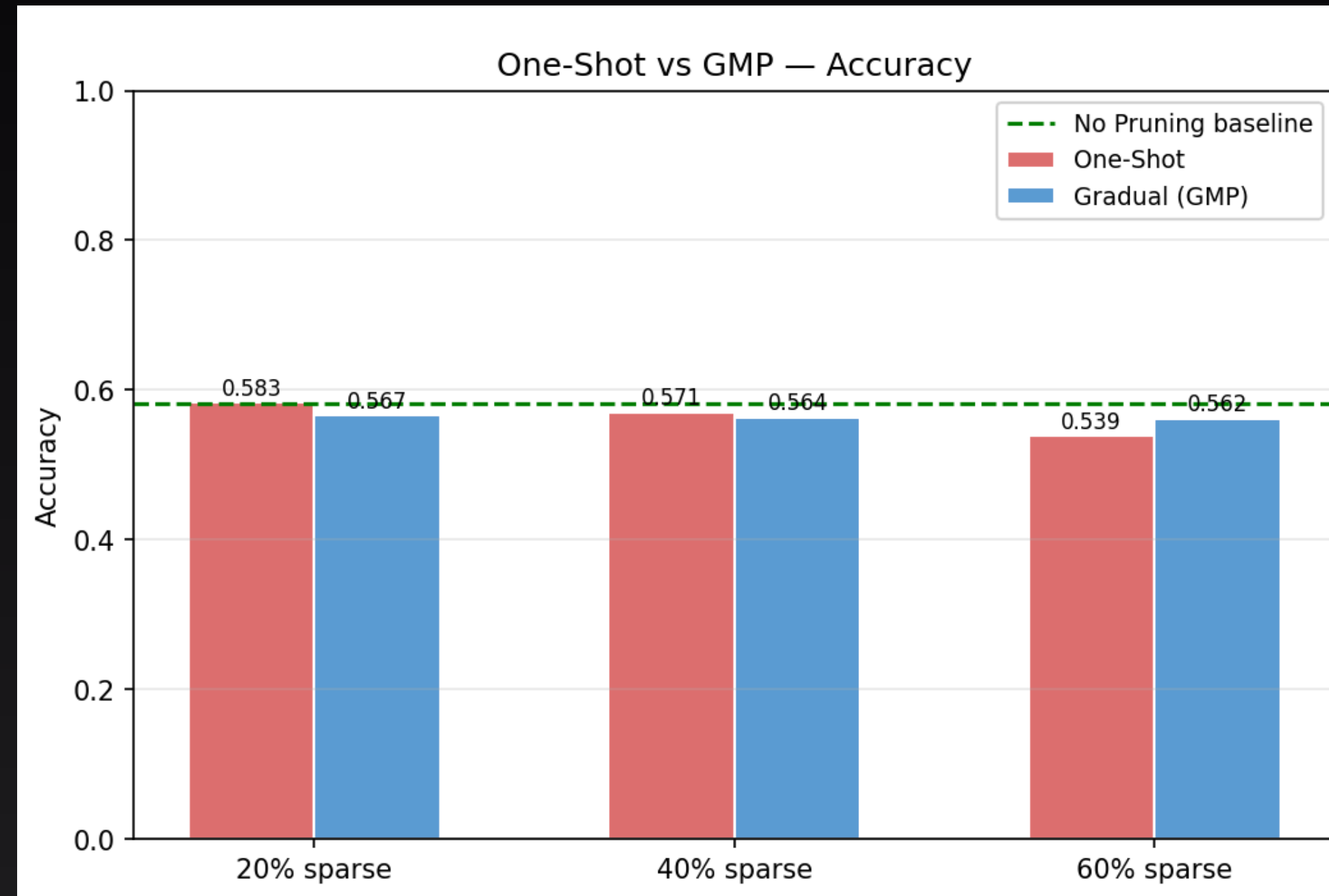
IMPROVEMENTS DONE:-

- Best-model checkpoint saving
- Class-balanced loss
- Longer training
- Better optimizer
- Learning-rate scheduling

Improvement 1: Gradual Magnitude Pruning

CREMA - D DATASET

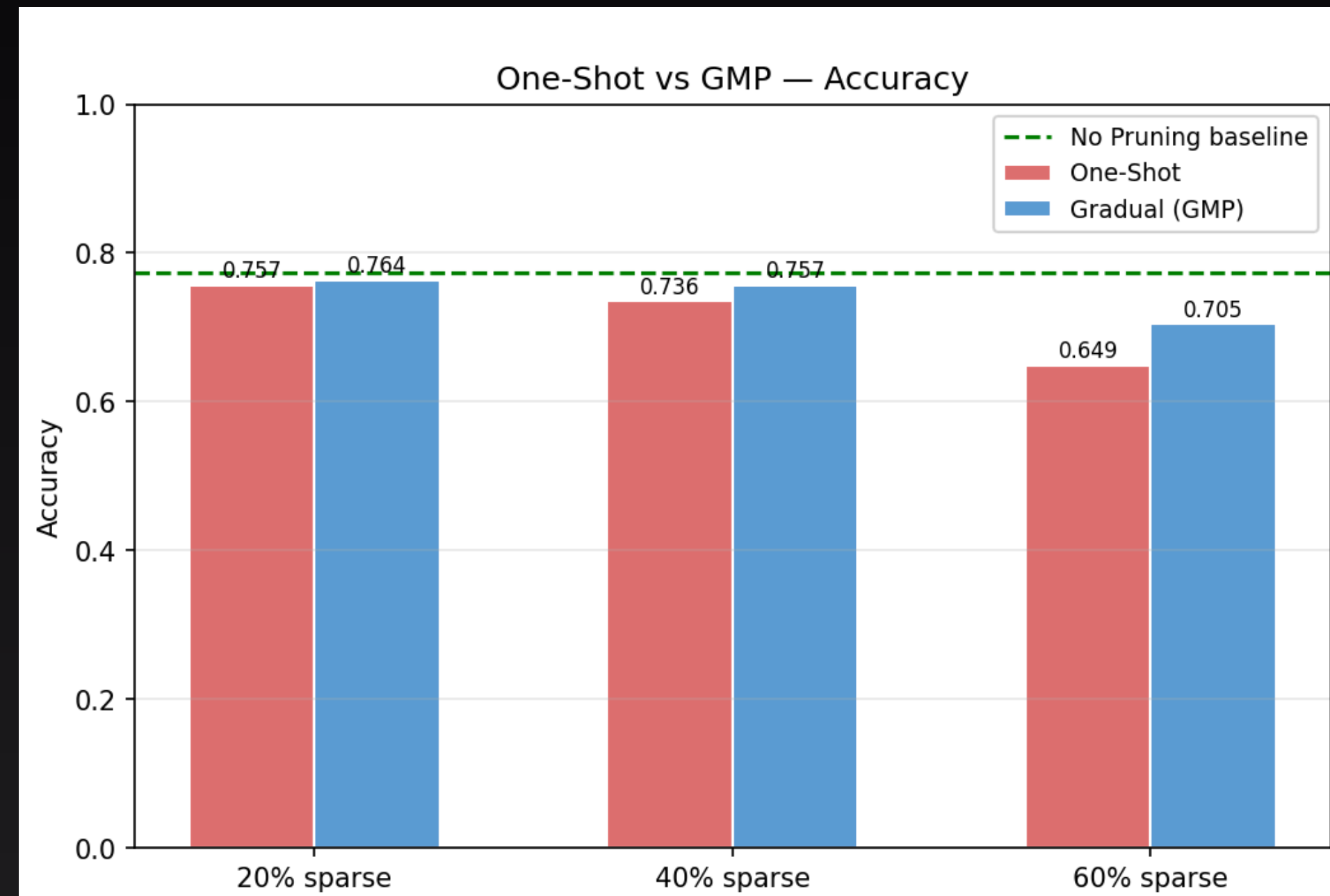
Dataset	Method	Accuracy	Macro F1
CREMA-D	One-Shot (60%)	53.9%	53.6%
CREMA-D	GMP (60%)	56.2%	55.9%



Improvement 1: Gradual Magnitude Pruning

RAVDESS DATASET

Dataset	Method	Accuracy	Macro F1
RAVDESS	One-Shot (60%)	64.9%	63.0%
RAVDESS	GMP (60%)	70.5%	69.9%



Improvement 2: Combined Pruning + Quantization

Dataset	Baseline	GMP + INT8
CREMA-D	58.2%	56.5%
RAVDESS	77.4%	75.0%

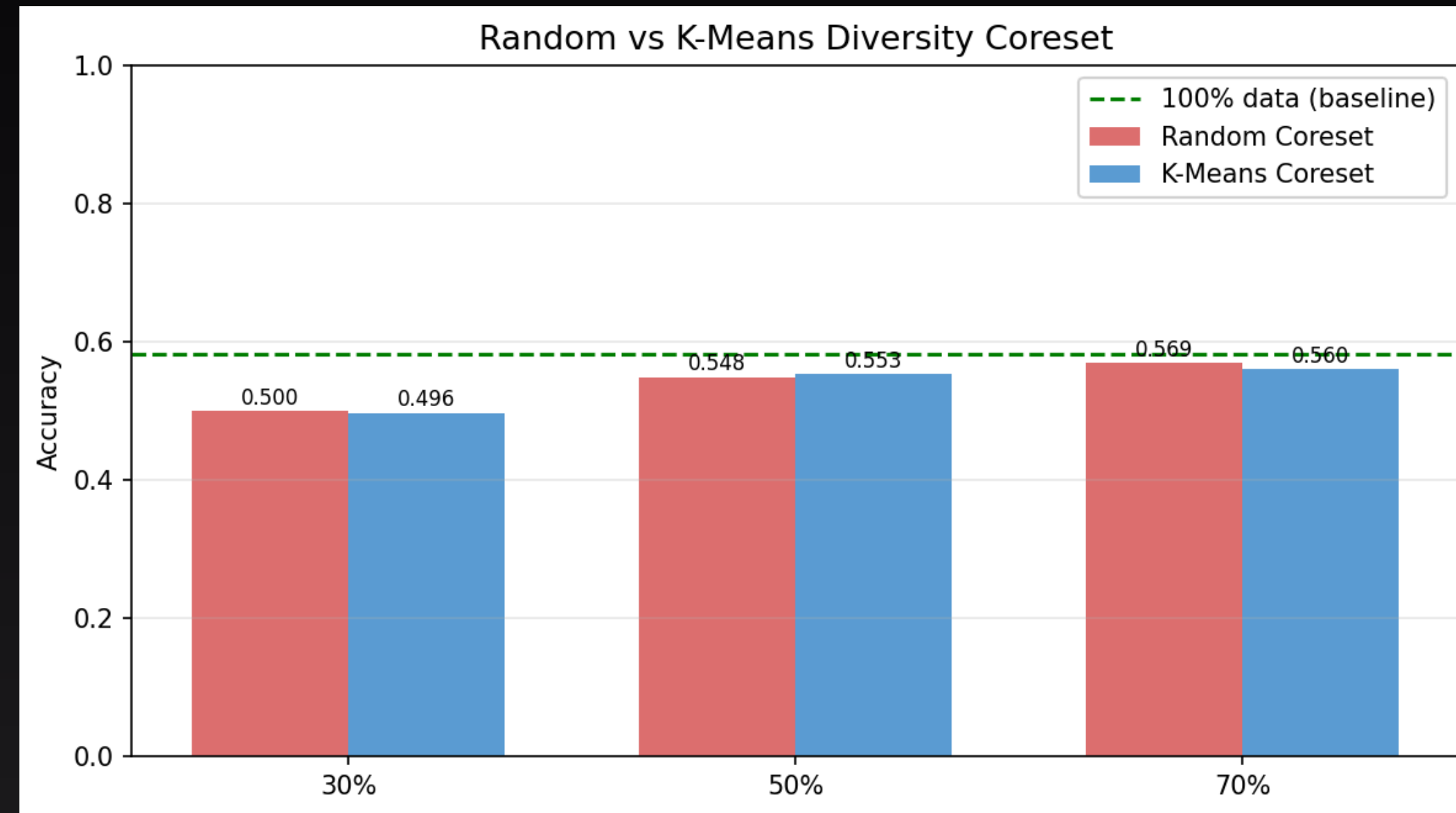
Model	Compression Ratio
Baseline	1×
GMP + INT8	4×

4× smaller model with minimal accuracy loss.

Improvement 3: K-Means Diversity-Based Coresets

CREMA - D DATASET

Dataset	Coreset Size	Random	K-Means
CREMA-D	50%	54.8%	55.3%

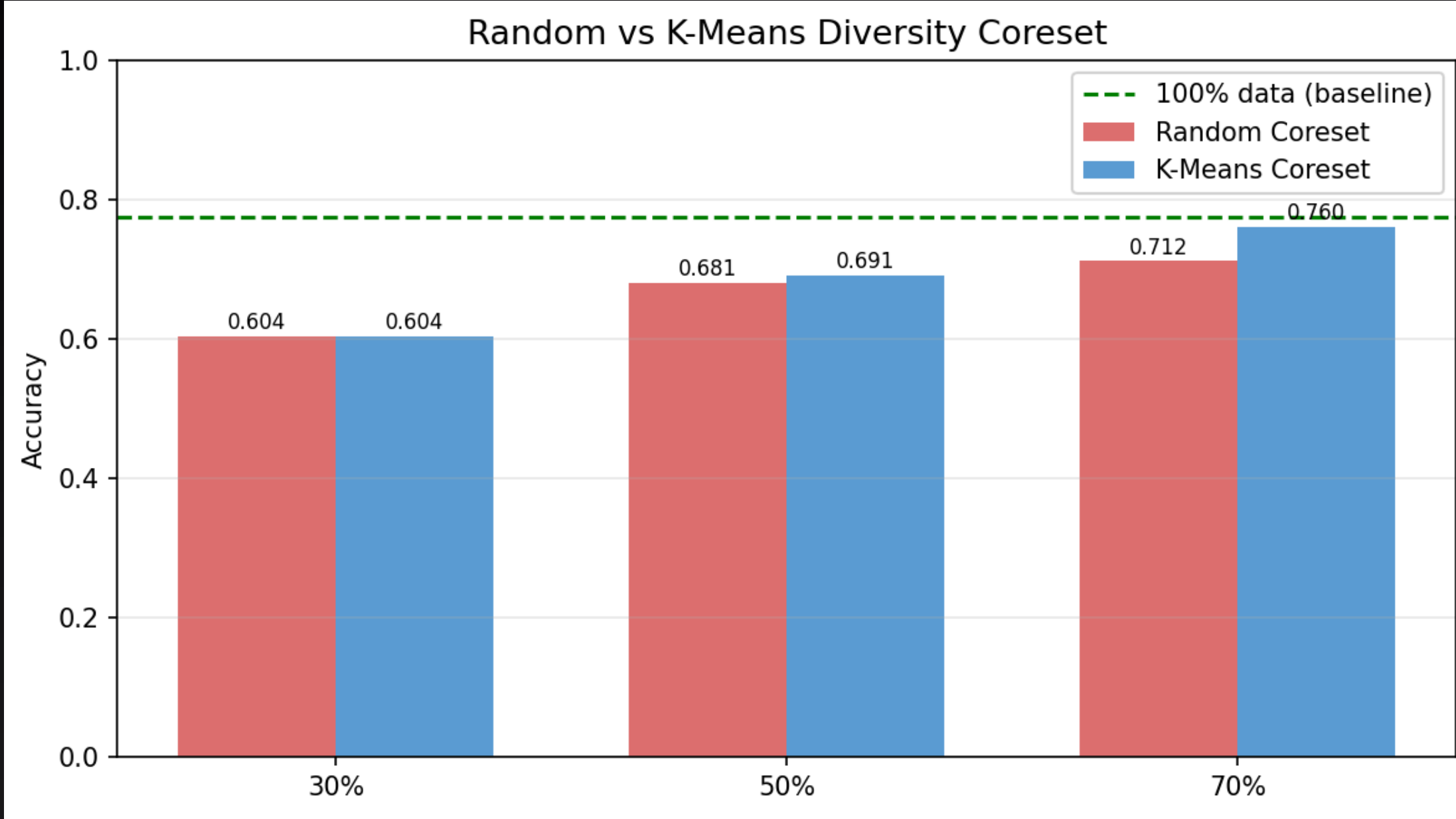


Diversity-based sampling preserves information better than random sampling.

Improvement 3: K-Means Diversity-Based Coresets

RAVDESS DATASET

Dataset	Coreset Size	Random	K-Means
RAVDESS	50%	68.1%	69.1%
RAVDESS	70%	71.2%	76.0%

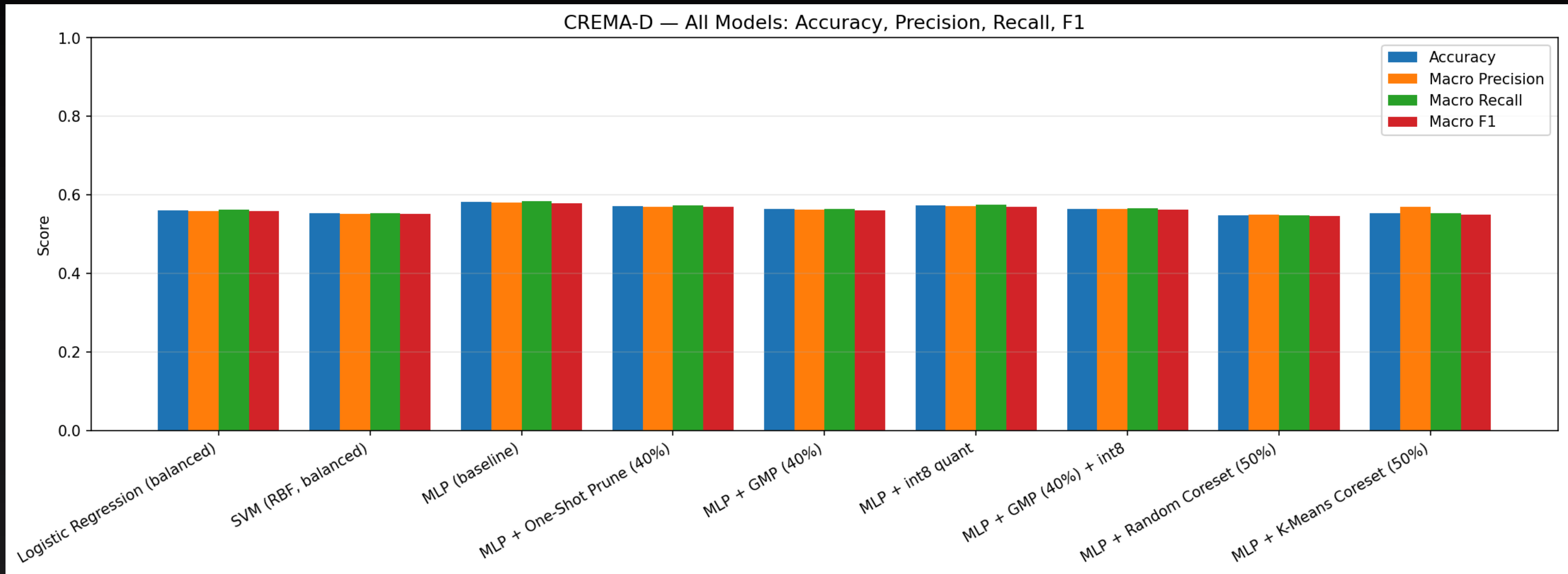


Diversity-based sampling preserves information better than random sampling.

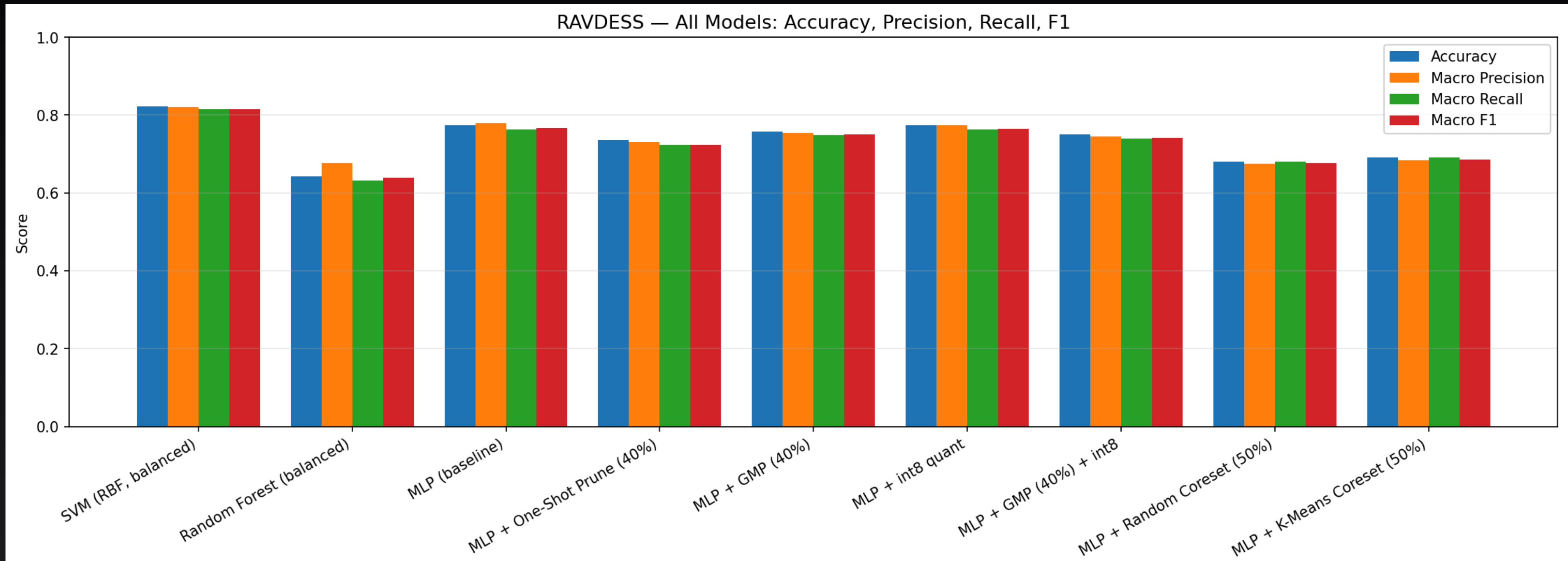
Comparison with Existing Research

Dataset	Study	Metric	Paper	Our Work
CREMA-D	MDPI Electronics 2022	Accuracy	53.66%	58.23%
RAVDESS	wav2vec 2.0 Study	Recall	77.5%	81.2%

SUMMARY - CREMA-D



SUMMARY – RAVDESS



CONCLUSION

GMP

- Better pruning robustness

Quantization

- 4× compression

K-Means

- Up to +4.8% accuracy improvement